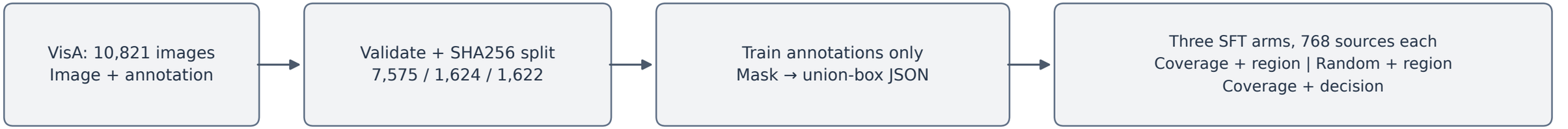
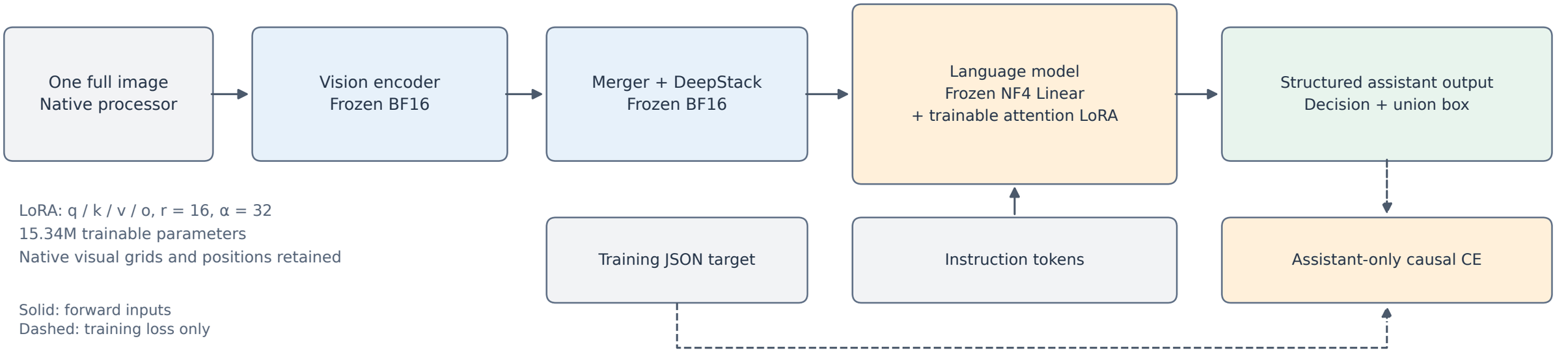
(a) Source-disjoint instruction construction

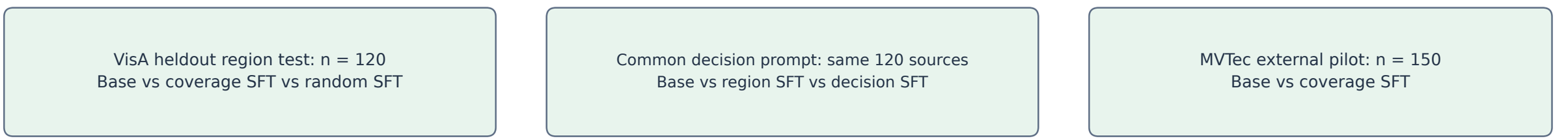


All source partitions fixed before instruction sampling; heldout annotations are used for scoring.

(b) Native Qwen3-VL with language-attention QLoRA



(c) Fixed-budget tests and controlled ablations



Single seed · 96 updates × 8 images per arm · max 524,288 pixels · greedy decoding · paired source bootstrap